HEMOSTASIS AND ROLE OF THROMBOCYTES

LEARNING OBJECTIVES

At the end of the lecture, the student should be able to:

- Define Hemostasis.
- Delineate the process of hemostasis that restricts blood loss when vessels are damaged.
- Give a picture of the structure of platelets.
- Describe the Role of platelets in hemostasis.
- Give the Abnormalities of Hemostasis.

HEMOSTASIS

- Most significant maintenance systems of human bodily homeostasis.

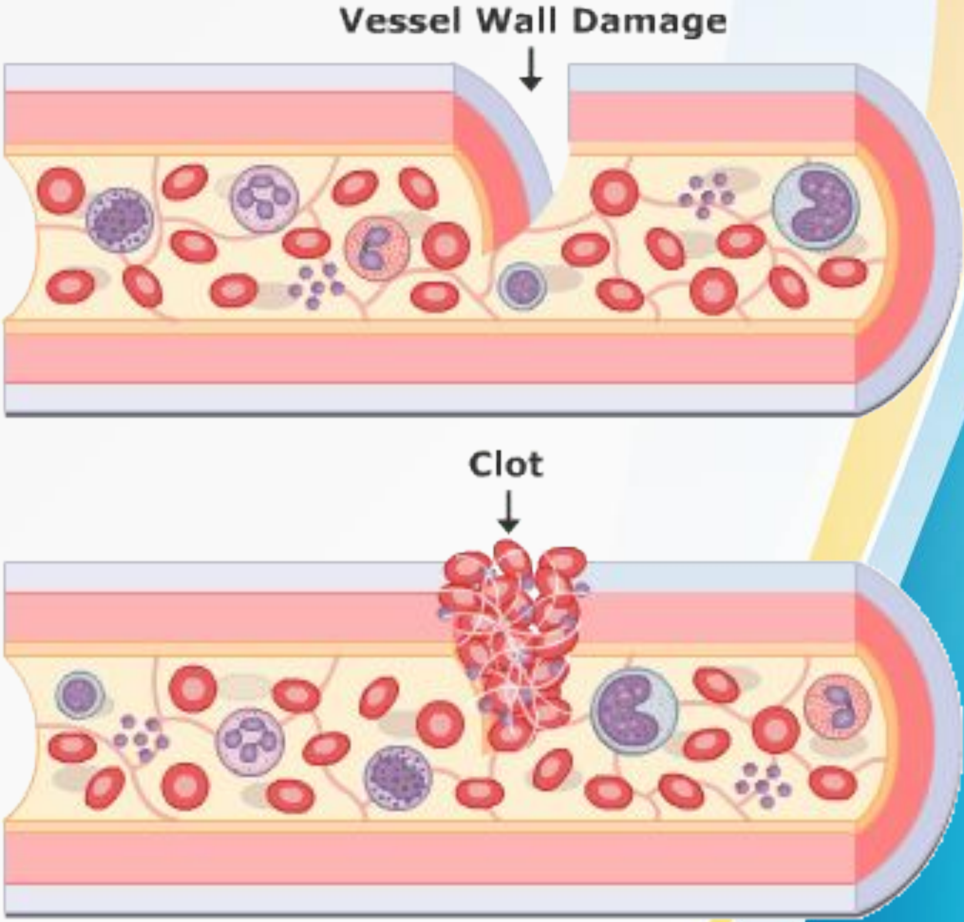
- Hemostasis plays two roles in the organism, namely:

- To provide blood flow through blood vessels, i.e., to maintain the liquid state of circulating blood and
- To prevent bleeding that results from blood vessel damage.

HEMOSTASIS

Definition:

The process of forming clots in walls of damaged blood vessels and preventing blood loss while maintaining blood in a fluid state within the vascular system.



Hemostasis is a complex system that includes the participation of several factors:

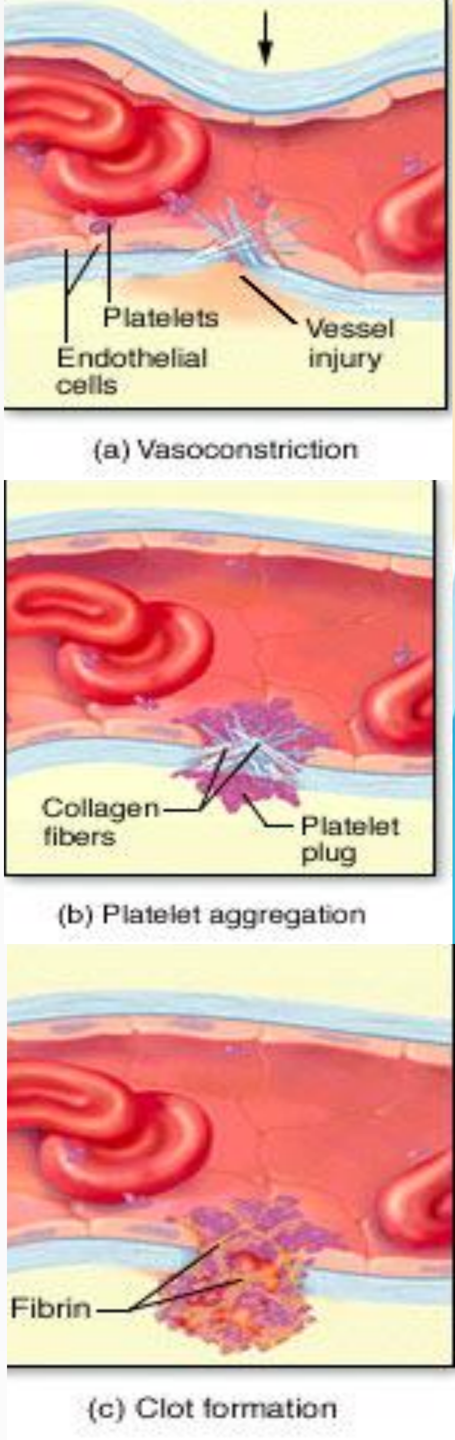
- blood vessel endothelium.
- platelets.
- blood coagulation.
- fibrinolytic process.
- coagulation inhibitors.

RESPONSE TO INJURY

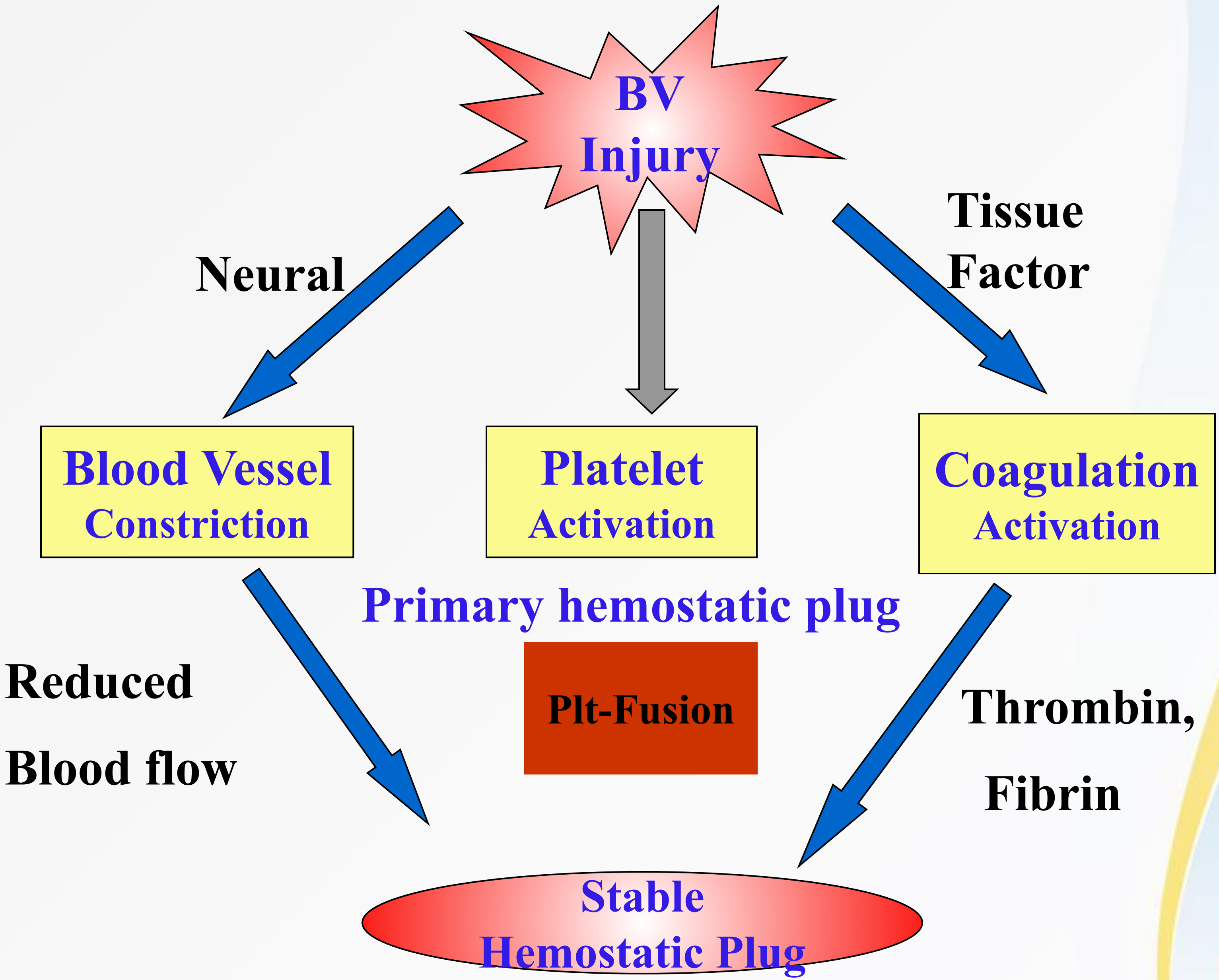
When blood vessel is transected, injury initiates a series of events that lead to hemostasis.

Hemostasis is achieved by:

- Vascular constriction,
- Formation of a platelet plug,
- Formation of a blood clot , and
- Eventual growth of fibrous tissue into the blood clot.



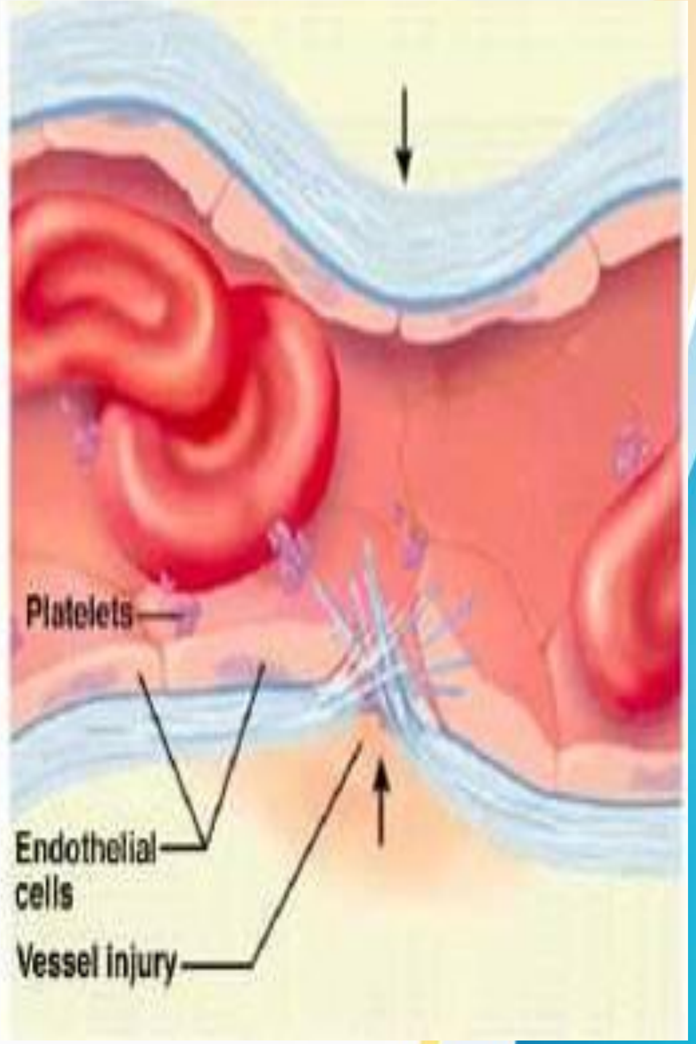
(a) Vasoconstriction
(b) Platelet aggregation
(c) Clot formation


BV
Injury
Neural
Tissue
Factor
Blood Vessel
Constriction
Platelet
Activation
Coagulation
Activation
Primary hemostatic plug
Plt-Fusion
Reduced
Blood flow
Thrombin,
Fibrin
Stable
Hemostatic Plug

STEPS OF HEMOSTASIS

VASOCONSTRICTION :

- Role of the blood vessel in haemostasis is the constriction of an injured blood vessel -vasoconstriction.
- This lasts less than a minute.
- is a reflex action that is prolonged by serotonin.



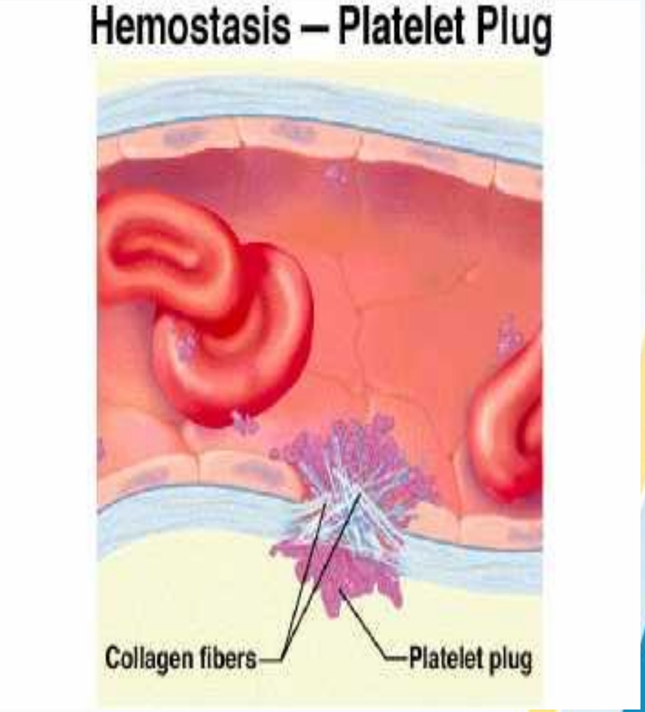
The contraction results from:

- local myogenic spasm,
- local autacoid factors from the traumatized tissues and platelets,
- nervous reflexes.

PRIMARY HEMOSTASIS

- Platelets adhere to exposed extracellular matrix by binding to von Willebrand factor (vWF) and are activated, undergoing a shape change and granule release.

- Released ADP and thromboxane A2 lead to further platelet aggregation (via binding of fibrinogen to platelet receptors) to form the **primary hemostatic plug.**

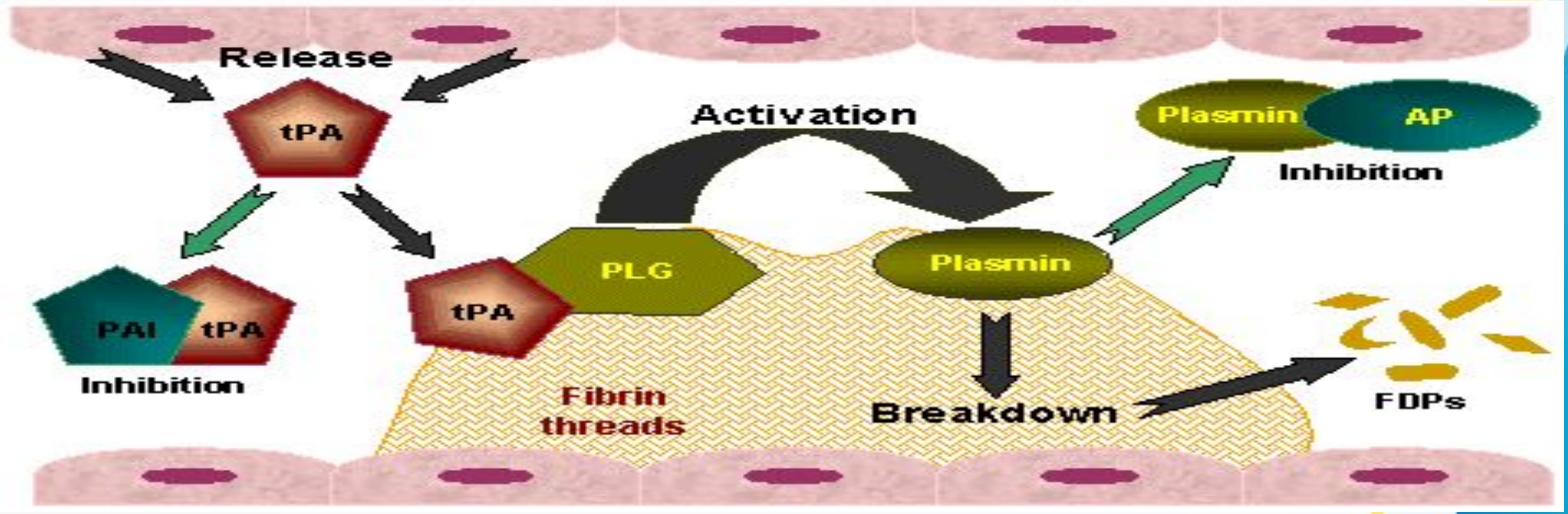


SECONDARY HEMOSTASIS

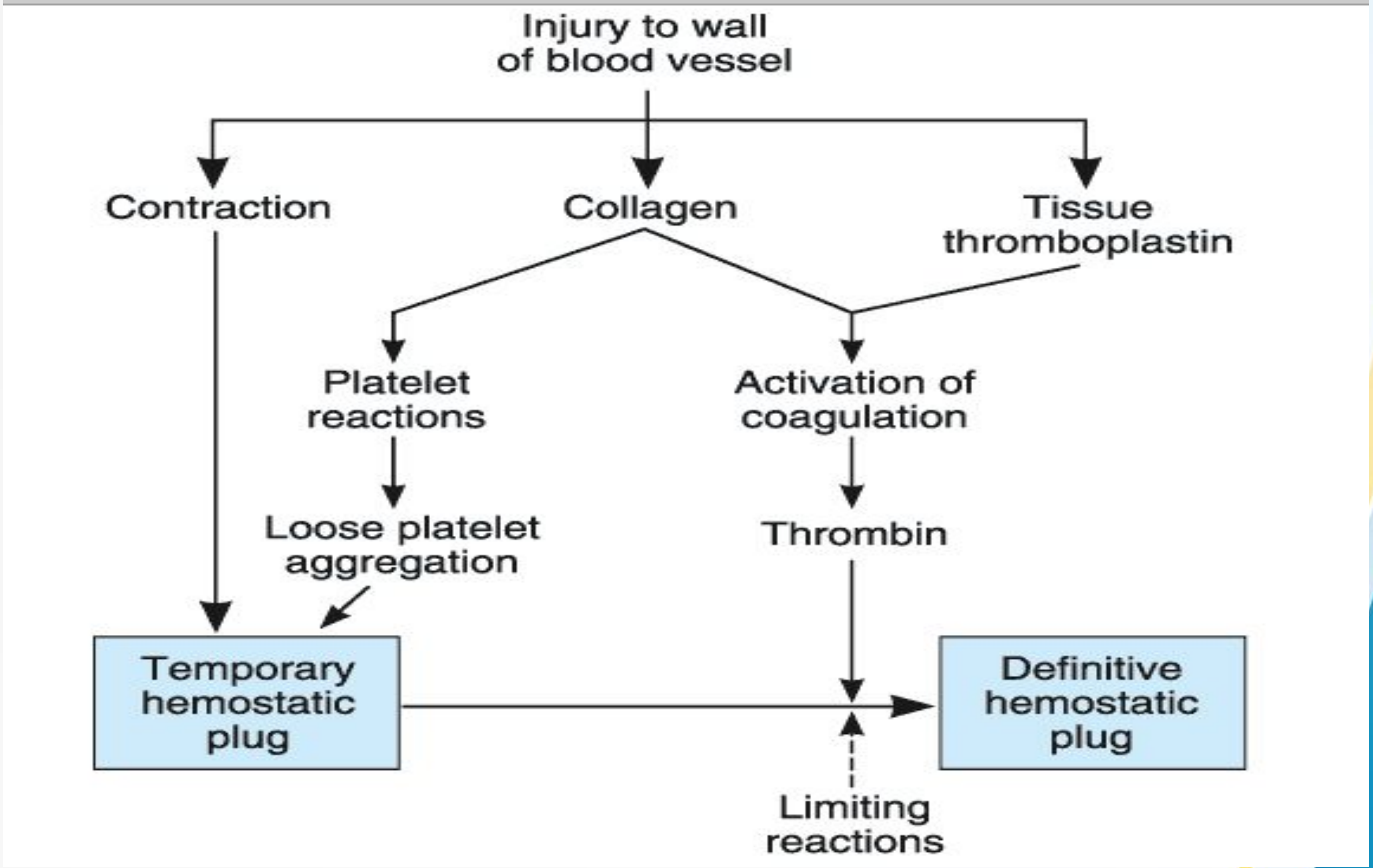
Local activation of coagulation cascade (involving tissue factor & platelet phospholipids) results in fibrin polymerization, ‘cementing’ the platelets into a definitive secondary hemostatic plug.

ANTITHROMBOTIC COUNTER REGULATION

Release of t-PA (tissue plasminogen activator), a fibrinolytic product and thrombomodulin, limit the hemostatic process to the site of injury.



SUMMARY OF HEMOSTASIS



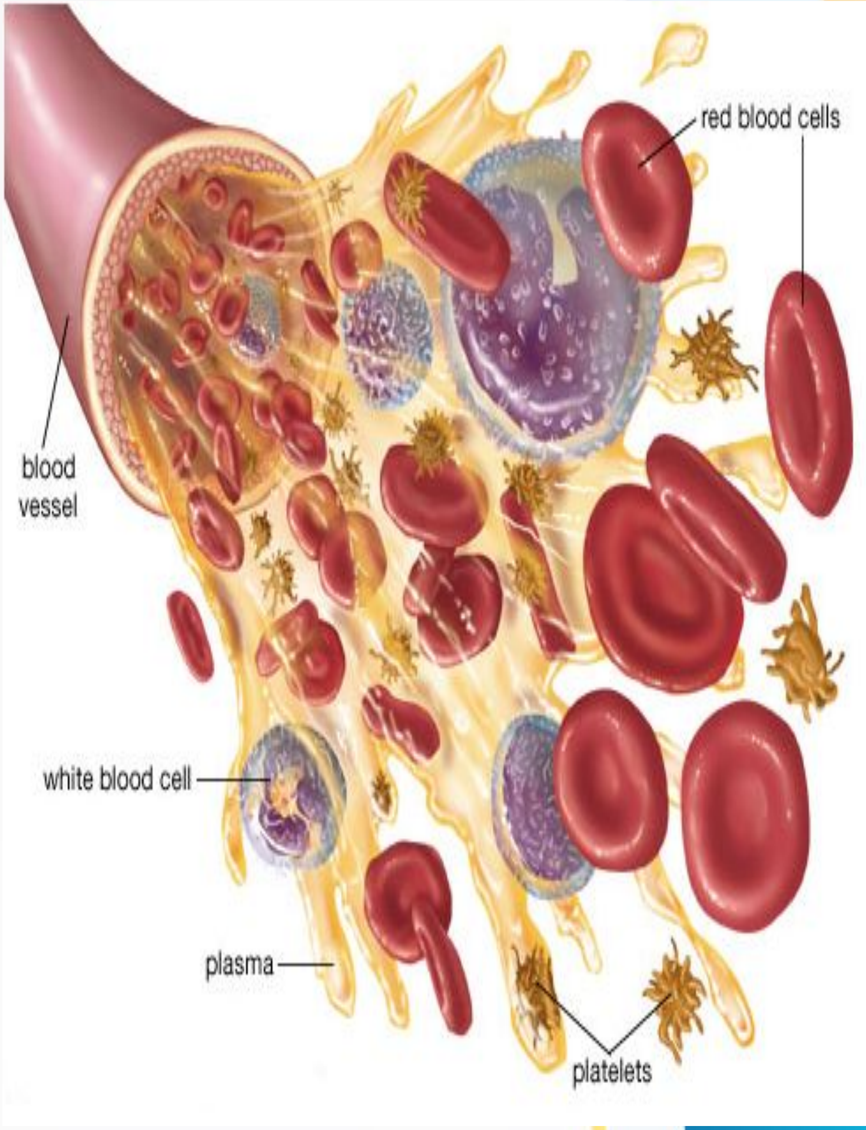
The dashed arrow indicates inhibition.

BLOOD CLOTTING FACTORS

Factor[a]	Names
I	Fibrinogen
II	Prothrombin
III	Thromboplastin
IV	Calcium
V	Proaccelerin, labile factor, accelerator globulin
VII	Proconvertin, SPCA, stable factor
VIII	Antihemophilic factor (AHF), antihemophilic factor A, antihemophilic globulin (AHG)
IX	Plasma thromboplastic component (PTC), Christmas factor, antihemophilic factor B
X	Stuart-Prower factor
XI	Plasma thromboplastin antecedent (PTA), antihemophilic factor C
XII	Hageman factor, glass factor
XIII	Fibrin-stabilizing factor, Laki-Lorand factor
HMW-K	High-molecular-weight kininogen, Fitzgerald factor
Pre-K_a	Prekallikrein, Fletcher factor
Ka	Kallikrein

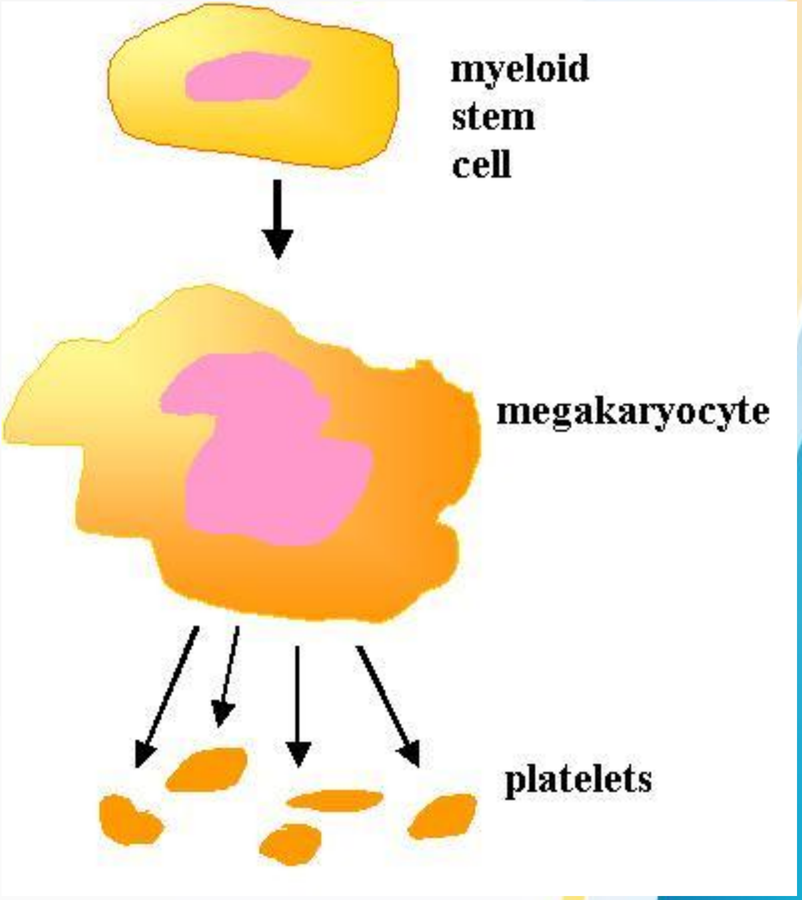
CHARACTERISTICS OF PLATELETS

- Also called *thrombocytes*
- Aggregate at sites of injury.
- Minute discs 1-4 micrometers in diameter.
- Formed in the bone marrow from *megakaryocytes*
- Normal concentration of platelets in blood is between 150,000 and 300,000 per microliter.
- Non-nucleated and cannot reproduce.
- Count= 300,000/µl.
- Half life= 4 days.



PLATELETS

- Megakaryocytes, giant cells in bone marrow, form platelets by pinching off bits of cytoplasm.
- 60%and 70% are in circulation
- Remainder in spleen.
- Splenectomy causes thrombocytosis.

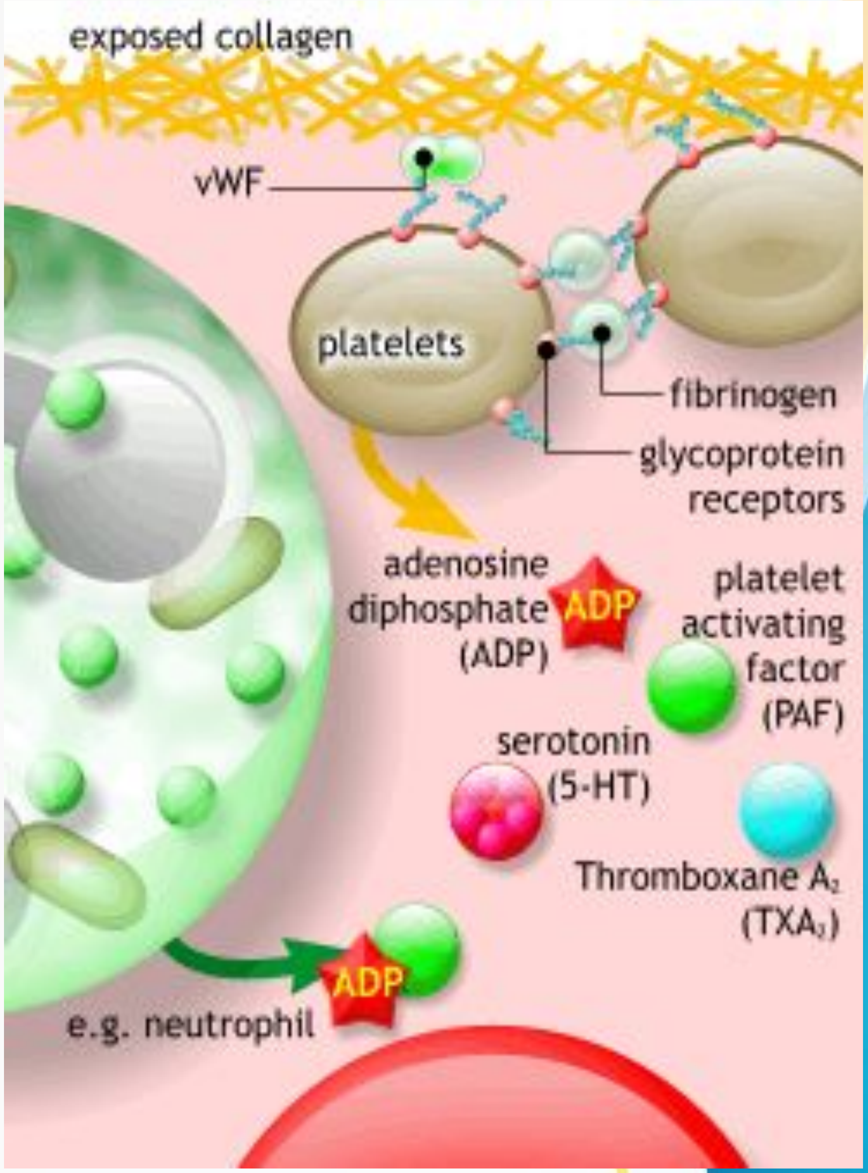


STRUCTURE OF PLATELETS

- Platelets have a ring of microtubules around their periphery and extensively invaginated membrane with an intricate canalicular system.

- Membranes contain receptors for collagen, ADP, vessel wall von Willebrand factor and fibrinogen.

- Cytoplasm contains actin, myosin, glycogen and lysosomes.

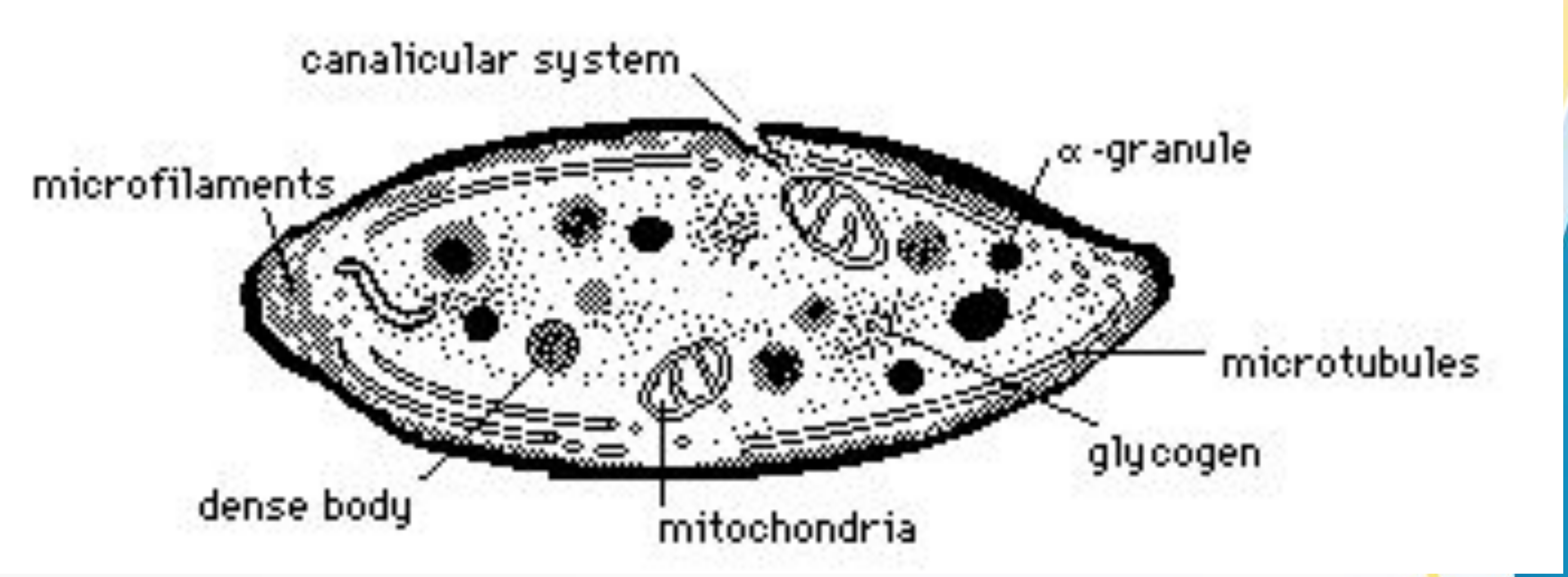


STRUCTURE OF PLATELETS

Two types of granules:

Dense granules: which contain the non protein substance e.g. serotonin and ADP

Alpha granules: containing secreted proteins e.g. PDGF.



- **Alpha granules contain:**

1. Clotting factors – fibrinogen, V and XIII
2. Platelet-derived growth factor
3. Vascular endothelial growth factor (VEGF)
4. Basic fibroblast growth factor (FGF)
5. Endostatin
6. Thrombospondin.

- **Dense granules**

Dense granules contain:

1. Nucleotides
2. Serotonin
3. Phospholipid
4. Calcium
5. Lysosomes.

- PDGF (platelet derived growth factor) is produced by macrophages and endothelial cells.
- PDGF stimulates wound healing and is a potent mitogen for vascular smooth muscle.
- Platelets contain von Willebrand factor. Its role in adhesion, regulates circulating levels of Factor VIII.
- Platelet production is regulated by colony stimulating factors for production of megakaryocytes.
- Thrombopoietin facilitates megakaryocytes maturation.

PROPERTIES OF PLATELETS

Platelets have three important properties (three ‘A’s):

- 1. Adhesiveness
- 2. Aggregation
- 3. Agglutination.

ADHESIVENESS

Adhesiveness is the property of sticking to a rough surface. During injury of blood vessel, endothelium is damaged and the subendothelial collagen is exposed. While coming in contact with collagen, platelets are activated and adhere to collagen.

Adhesion of platelets involves interaction between von Willebrand factor secreted by damaged endothelium and a receptor protein called glycoprotein Ib situated on the surface of platelet membrane.

Other factors which accelerate adhesiveness

- are collagen, thrombin, ADP, Thromboxane A2, calcium ions, P-selectin and vitronectin.

AGGREGATION

- Aggregation is the grouping of platelets. Adhesion is followed by activation of more number of platelets by substances released from dense granules of platelets. During activation, the platelets change their shape with elongation of long filamentous pseudopodia which are called processes or filopodia .
Filopodia help the platelets aggregate together. Activation and aggregation of platelets is accelerated by ADP, thromboxane A2 and platelet-activating factor (PTA: cytokine secreted by neutrophils and monocytes

AGGLUTINATION

- Agglutination is the clumping together of platelets. Aggregated platelets are agglutinated by the actions of some platelet agglutinins and platelet-activating factor.

FUNCTIONS OF PLATELETS

- Normally, platelets are inactive and execute their actions only when activated. Activated platelets immediately release many substances. This process is known as platelet release reaction. Functions of platelets are carried out by these substances.Functions of platelets are:

 1. **ROLE IN BLOOD CLOTTING**

Platelets are responsible for the formation of intrinsic prothrombin activator. This substance is responsible for the onset of blood clotting .

 2. ROLE IN CLOT RETRACTION

In the blood clot, blood cells including platelets are entrapped in between the fibrin threads. Cytoplasm of platelets contains the contractile proteins, namely actin, myosin and thrombosthenin, which are responsible for clot retraction.

3. ROLE IN PREVENTION OF BLOOD LOSS (HEMOSTASIS)

- Platelets accelerate the hemostasis by three ways:
- i. Platelets secrete 5-HT, which causes the constriction of blood vessels.
- ii. Due to the adhesive property, the platelets seal
- the damage in blood vessels like capillaries.
- iii. By formation of temporary plug, the platelets
- seal the damage in blood vessels.

4. ROLE IN REPAIR OF RUPTURED BLOOD VESSEL

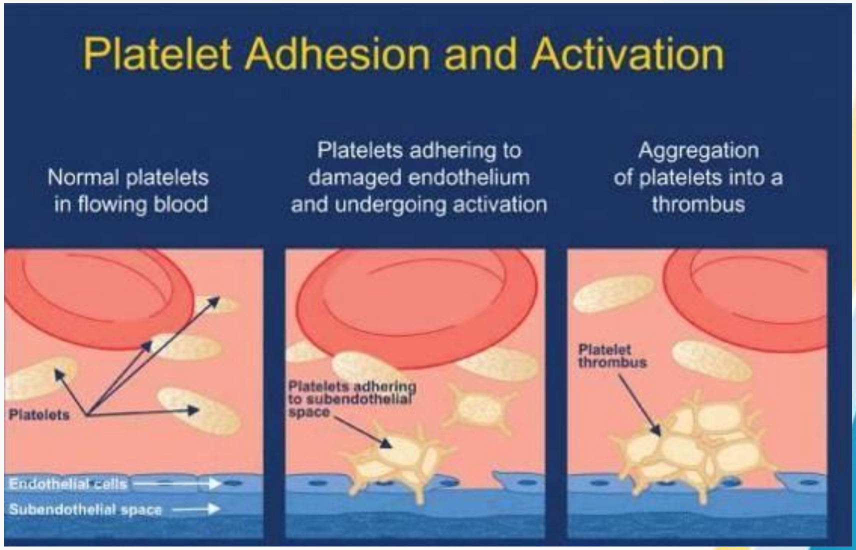
- Platelet-derived growth factor (PDGF) formed in cyto -
- plasm of platelets is useful for the repair of the endothelium and other structures of the ruptured blood vessels.

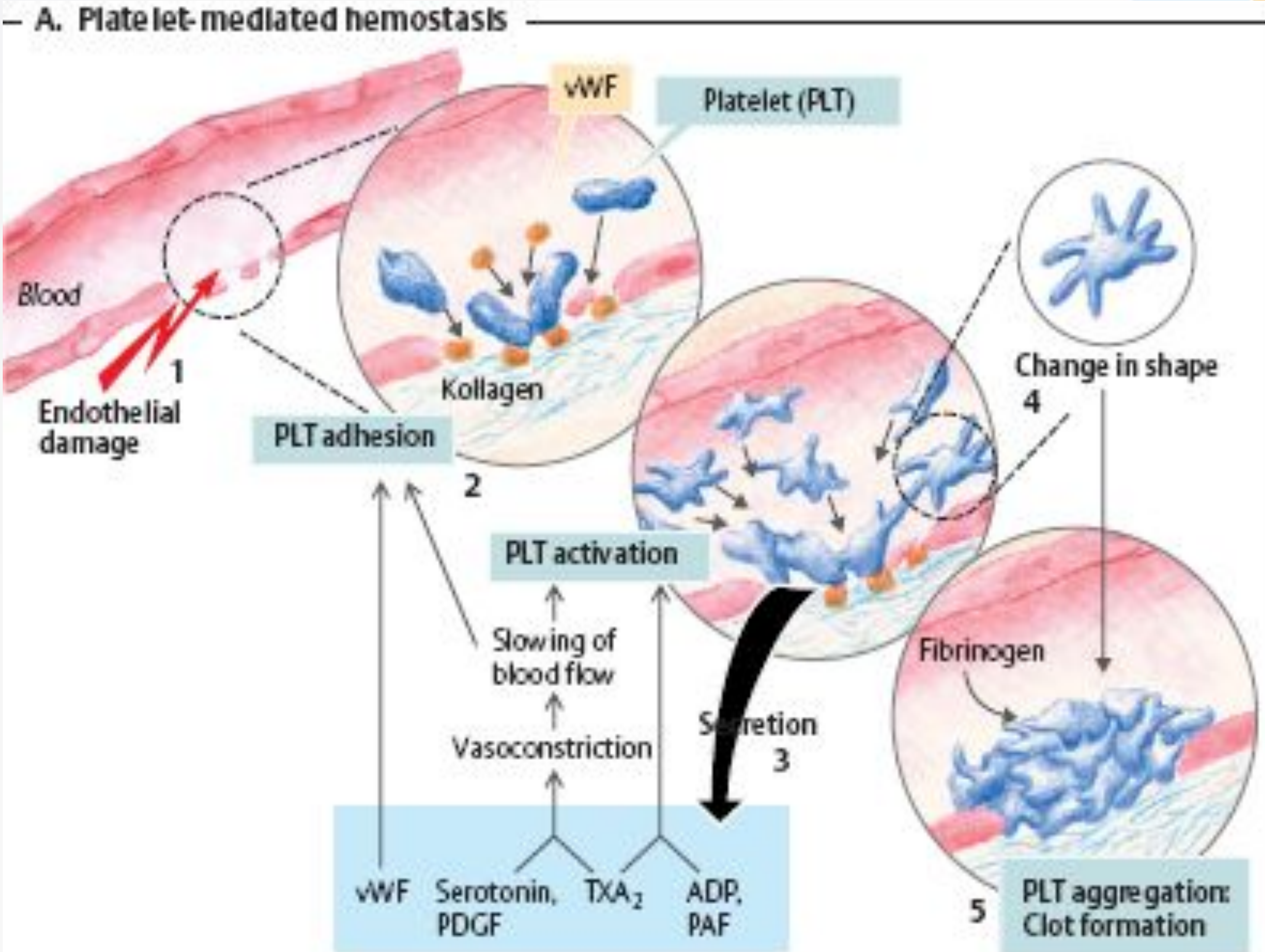
- **5. ROLE IN DEFENSE MECHANISM**
- By the property of agglutination, platelets encircle the
- foreign bodies and destroy them.

ROLE OF PLATELETS IN HEMOSTASIS

- Endothelial injury leads to platelet aggregation.
- Platelet adhesion.
- Platelet activation.
- Secretion.
- Change in shape.
- Platelet aggregation:

  clot formation

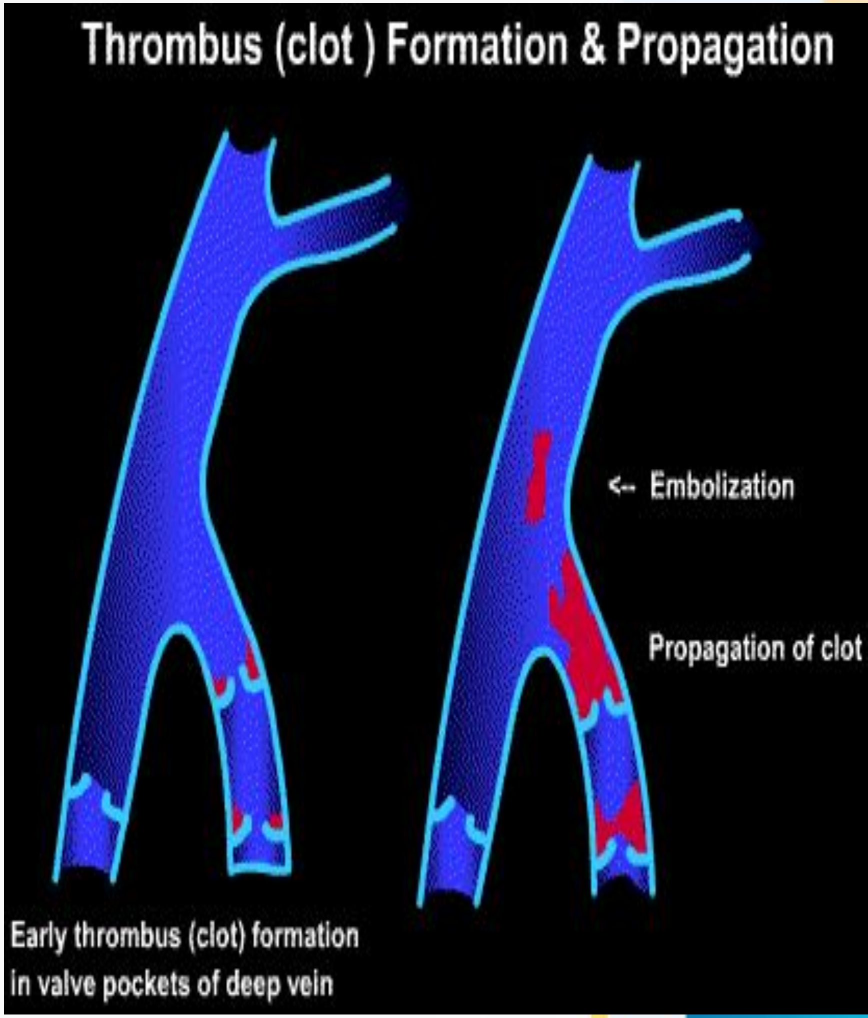



A. Platelet-mediated hemostasis
vWF
Platelet (PLT)
Blood
1
Endothelial damage
Kollagen
PLT adhesion
2
Change in shape
4
PLT activation
Slowing of blood flow
Vasoconstriction
Secretion
3
Fibrinogen
vWF
Serotonin, PDGF
TXA2
ADP, PAF
5
PLT aggregation: Clot formation

ABNORMALITIES OF HEMOSTASIS

THROMBOSIS

- Formation of clots inside blood vessels
- occur where blood flow is sluggish because the slow flow permits activated clotting factors to accumulate.
- frequently occlude arterial supply to the organs in which they form
- bits of thrombus **(emboli)** sometimes break off & travel in the bloodstream damaging other organs



PURPURA AND ITS TYPES

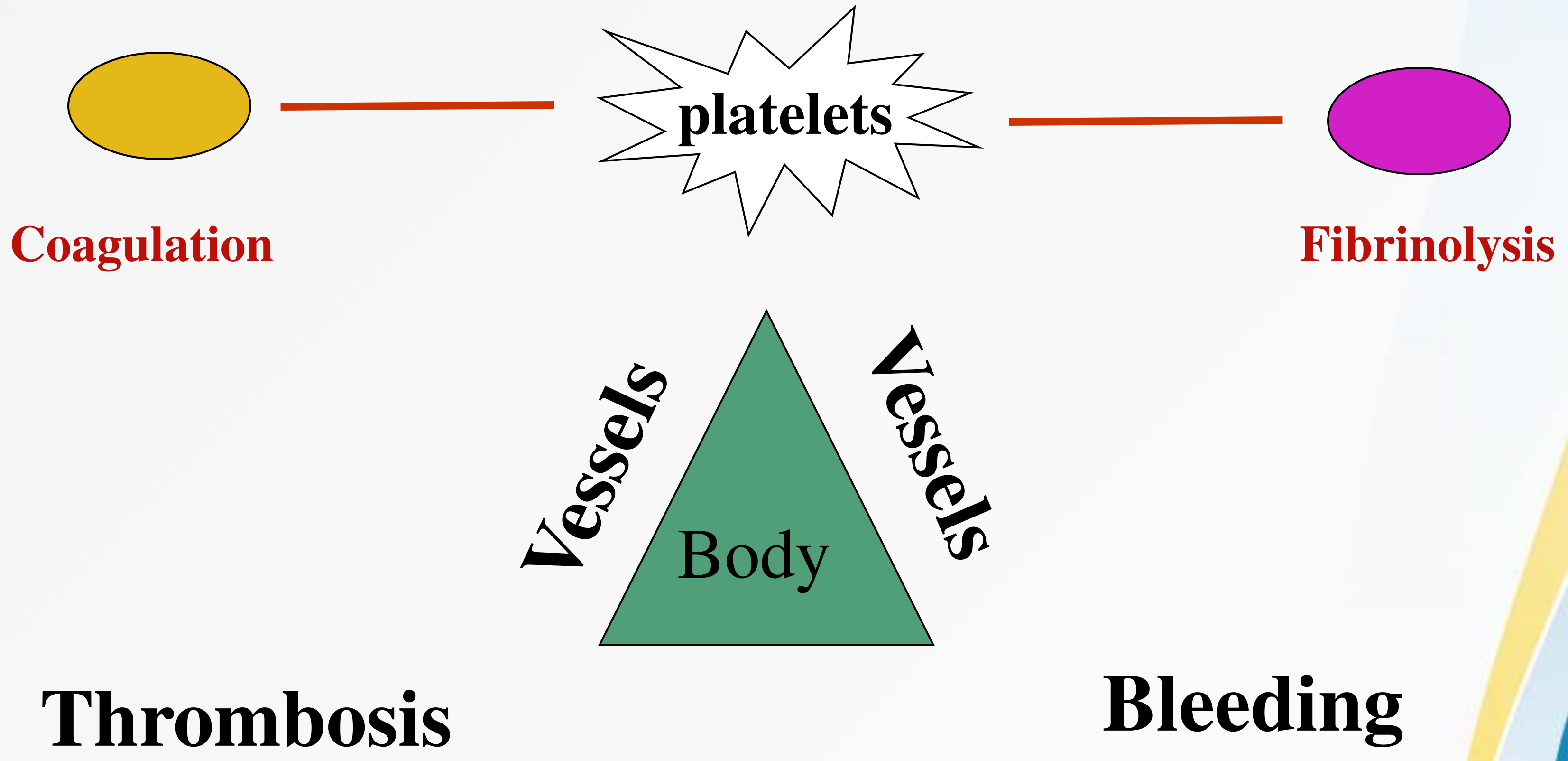
- **<u>THROMBOCYTOPENIA PURPURA</u>**: When platelets count is low, clot retraction is deficient and there is poor constriction of ruptured vessels leading to purpura.

- **<u>THROMBASTHENIC PURPURA:</u>**
  Platelet count is normal but abnormal circulating platelets.

ANTI CLOTTING MECHANISMS

- **Endothelium:** There is a balance between endothelial antithrombotic and prothrombotic activities.
- **Antithrombotic properties-prostacyclin and nitric oxide** are potent vasodilators and inhibitors of platelet aggregation.
- **Anticoagulant effects:** Thrombomodulin heparin –like molecules.
- **Antithrombin III :** Inactivates thrombin and factor Xa.
- **Fibrinolytic properties:** Plasminogen activator

IMPORTANCE OF BALANCE IN HEMOSTASIS



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